2022B F=ma Exam: Problem 22

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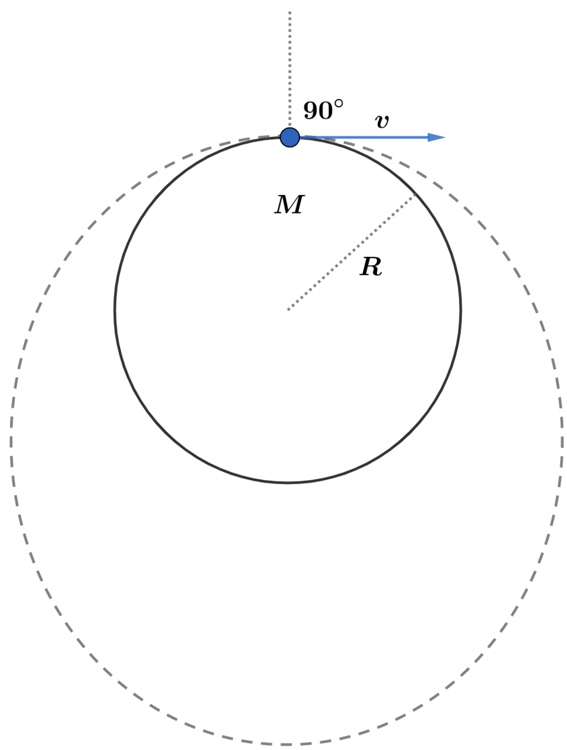

Note that the speed for a circular orbit right above the surface of the planet is

$$v_{\text{cir}} = \sqrt{\frac{GM}{R}}$$

and the escape velocity is

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

We have $v_{\text{cir}} < v < v_{\text{esc}}$ so the particle must have an elliptical orbit. If the particle never collides with the planet, then the starting point must be the closet point i.e. the perigee. Thus, we must have $\theta = 90°$ so the answer is $\boxed{\text{D}}$.